

Unit Test Frameworks

Assignment 7: Page Object Model (POM) Restructure

1. Problem Statement: -

Take your working scripts from Tier 1 and restructure them into a Page Object Model design pattern. Create separate page classes (e.g., LoginPage) containing only locator definitions and UI methods. Keep actual test assertions completely separate from the locators.

2. Objective: -

To implement the Page Object Model (POM) design pattern to enhance code reusability, maintainability, and readability by separating UI locators from test logic.

3. Tools, Software, and Concepts Used: -

Python 3.x, Selenium WebDriver, Object-Oriented Programming (OOP).

4. Source Code / Implementation: -

```
from selenium.webdriver.common.by import By
class LoginPage:
    def __init__(self, driver):
        self.driver = driver
        self.username_input = (By.ID, "user-name")
        self.password_input = (By.NAME, "password")
        self.login_button = (By.XPATH, "//*[@id='login-button']")
    def enter_username(self, username):
        self.driver.find_element(*self.username_input).send_keys(username)
    def enter_password(self, password):
        self.driver.find_element(*self.password_input).send_keys(password)
    def click_login(self):
        self.driver.find_element(*self.login_button).click()
from selenium import webdriver
import time
driver = webdriver.Chrome()
driver.get("https://www.saucedemo.com/")
login_page = LoginPage(driver)
login_page.enter_username("standard_user")
login_page.enter_password("secret_sauce")
time.sleep(5)
login_page.click_login()
assert "/inventory.html" in driver.current_url
```

```
print("POM Login Successful!")
time.sleep(5)
driver.quit()
```

5. Output & Screenshots: -

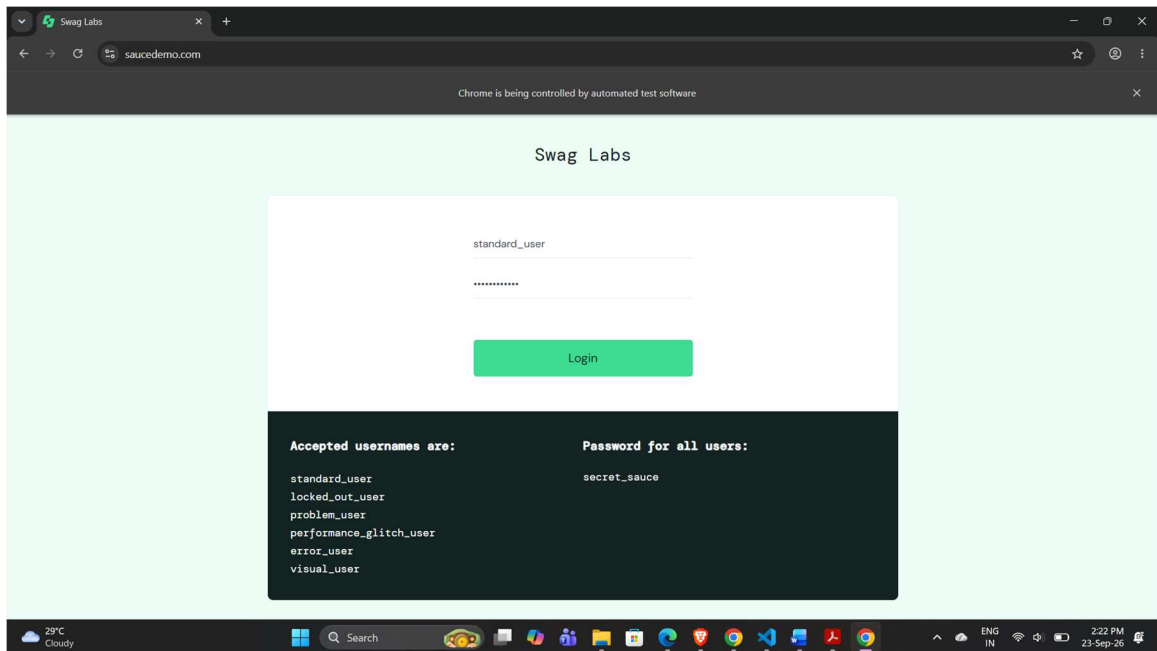


Fig.1: Successfully password is filling up the box

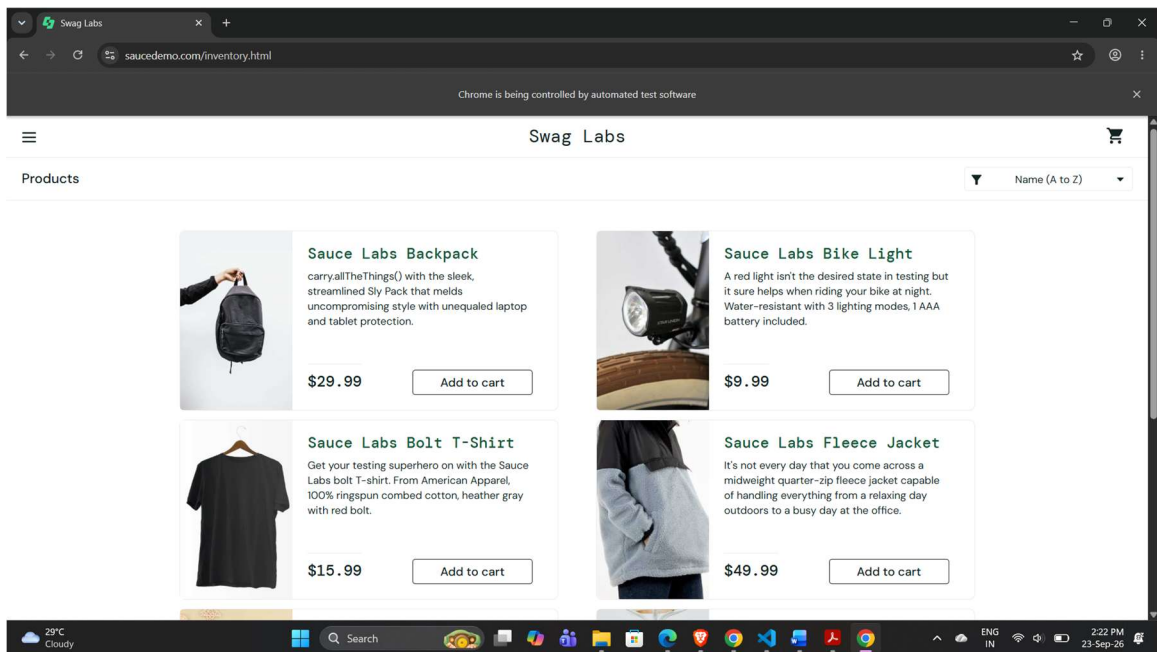


Fig.2: Successfully login the page

6. Result, Observation, and Conclusion: -

- Result: The login script was successfully refactored into a page class and a test execution script.
- Observation: POM significantly reduces code duplication. If a locator changes in the UI, it only needs to be updated in one place (the Page Class), rather than across multiple test scripts.
- Conclusion: Separating UI definitions from test assertions is a fundamental best practice for scalable automation frameworks.